

# Talk-to-Data Delivery Blueprint — Phase 4 Annexes: Design Architecture

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## 1 How to use this annex

This annex pack contains example tools, templates and checklists supporting the **Phase 4 Design Architecture and Orchestration Guide**. It is intended as practical inspiration for facilitation, documentation and delivery, not as a mandatory to-do list.

The main guide explains the delivery logic and decisions required during Phase 4. This annex provides reusable working material that teams can adapt depending on the delivery stage, business risk, data maturity, technology stack, security requirements, model choices and governance requirements.

Teams should use only the templates that add value. For a narrow POC, many annex items may be simplified or skipped. For an MVP, pilot or production path, more of the material may be needed to support traceability, testing, security, cost control, ownership and operational handover.

The first three sections are cross-cutting. Model patterns, cost scenarios and evaluation / model-switching support Activities 5.3, 5.4, 5.9 and 5.10 in the main guide. The activity worksheets in Section 5 support specific Phase 4 design activities.

## 2 Model patterns

### 2.1 Model family and likely use

| Model family                   | Example models / technologies                                              | Likely use in T2D                                          | Strengths                             | Watch-outs                            |
|--------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------|---------------------------------------|
| Rules / deterministic logic    | SQL rules, policy engines, regex, validation services                      | Access checks, routing rules, thresholds, validation gates | Cheap, auditable, predictable         | Limited flexibility                   |
| Embedding model                | text-embedding models, voyage embeddings, bge, e5                          | Metadata search, semantic matching, example retrieval      | Low cost, fast, useful for grounding  | Retrieval quality must be tested      |
| Small / local / embedded model | GPT-4o mini, GPT-4.1 mini, Gemini Flash, Mistral Small, local 3B–8B models | Intent routing, metadata ranking, simple classification    | Cheap, fast, easier to deploy locally | May fail on ambiguity                 |
| Medium general model           | GPT-4.1, Claude Sonnet, Gemini Flash/Pro, Mistral Medium / Large           | Query drafting, clarification, answer shaping              | Good balance of cost and capability   | Needs strong grounding and validation |
| Strong reasoning model         | OpenAI reasoning models, Claude Opus, Gemini Pro reasoning models          | Complex SQL, ambiguity resolution, hard edge cases         | Better reasoning and robustness       | Higher cost and latency               |

| model                            | Gemini Pro reasoning models                                 | resolution, hard edge cases                                | robustness                                |                                                   |
|----------------------------------|-------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------|---------------------------------------------------|
| Self-hosted / private model      | Llama, Mistral open-weight, Qwen, private cloud deployments | Sensitive data, residency constraints, regulated use cases | More control over deployment and data     | Higher operating complexity                       |
| Fine-tuned / task-specific model | Fine-tuned small model, distilled model, classifier         | Repeated narrow tasks with stable patterns                 | Consistent, efficient for known workflows | Requires data, maintenance and regression testing |

## 2.2 Model use by orchestration task

| Task                      | Likely model pattern                     | Notes                                                         |
|---------------------------|------------------------------------------|---------------------------------------------------------------|
| Intent classification     | Rules, small model or medium model       | Use stronger models only if intent is ambiguous or high risk. |
| Metadata retrieval        | Embeddings plus small or medium model    | Retrieval quality matters more than model size.               |
| Metadata ranking          | Small or medium model                    | Must be evaluated against approved artefacts.                 |
| Clarification question    | Medium model                             | Should follow controlled clarification rules.                 |
| Query generation          | Medium or strong model                   | Stronger model may be justified for complex SQL.              |
| Query validation          | Rules plus model-assisted review         | Do not rely on the model alone for validation.                |
| Result interpretation     | Medium model                             | Must use query result, caveats and metric definitions.        |
| Answer generation         | Medium or strong model                   | Main risk is overstatement, missing caveats or wrong context. |
| Follow-up handling        | Medium or strong model                   | Risk increases when context is inherited across turns.        |
| Safety / refusal decision | Rules plus model-assisted classification | Refusal should be policy-led, not purely model-led.           |

## 3 Cost scenarios

### 3.1 Cost scenario worksheet

Cost should be estimated by orchestration step, not only as a total model bill. The same user interaction may trigger metadata retrieval, one or more model calls, query validation, execution, answer generation and logging.

| Scenario      | Initial request pattern                                                                                               | Follow-up pattern                                                 | Main cost drivers                                              |
|---------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------|
| Simple POC    | One metadata retrieval, one model call for query, one execution, one answer call                                      | Reuse context, limited new retrieval, one answer call             | Model calls, small query cost, basic logging                   |
| MVP           | Metadata retrieval, intent routing, query generation, validation, execution, answer generation                        | Context check, possible re-query, answer regeneration             | Model routing, validation calls, query execution, logs         |
| Complex pilot | Multi-step retrieval, stronger model for SQL, layered validation, execution, result interpretation, answer generation | Context inheritance, new filters, re-query, additional validation | Strong model use, warehouse cost, concurrency, trace retention |

### 3.2 Indicative cost structure

| Cost component          | Initial request             | Follow-up request                        | Notes                                       |
|-------------------------|-----------------------------|------------------------------------------|---------------------------------------------|
| Metadata retrieval      | Usually required            | May be reused or refreshed               | Use caching where safe.                     |
| Intent / routing model  | Usually required            | Usually required                         | Often suitable for a small model.           |
| Query generation model  | Required if query needed    | Required if follow-up needs new query    | Cost rises with schema/context size.        |
| Query validation        | Required before execution   | Required before re-execution             | Prefer deterministic checks where possible. |
| Query execution         | Required if data is queried | May be avoided if prior result is enough | Can dominate cost at scale.                 |
| Answer generation       | Required                    | Required                                 | Main quality and safety risk.               |
| Logging / trace storage | Required                    | Required                                 | Retention policy affects cost.              |
| Evaluation / regression | Periodic or batch           | Periodic or batch                        | Should be budgeted from MVP onward.         |

### 3.3 Indicative cost-per-interaction template

The values below are placeholders for planning, not vendor quotes. They should be replaced with organisation-specific pricing, model choice, token volumes, warehouse cost, retrieval pattern, logging retention and expected usage.

| Scenario      | Retrieval cost | Model-call cost | Query execution cost | Logging / trace cost | Indicative total per initial interaction | Indicative total per follow-up |
|---------------|----------------|-----------------|----------------------|----------------------|------------------------------------------|--------------------------------|
| Simple POC    | £0.001–£0.005  | £0.01–£0.05     | £0.001–£0.02         | <£0.001              | £0.01–£0.08                              | £0.005–£0.04                   |
| MVP           | £0.003–£0.02   | £0.03–£0.20     | £0.005–£0.10         | £0.001–£0.005        | £0.04–£0.30                              | £0.02–£0.18                    |
| Complex pilot | £0.01–£0.05    | £0.15–£1.00     | £0.05–£1.00+         | £0.005–£0.03         | £0.20–£2.00+                             | £0.10–£1.50+                   |

|      |       |  |  |  |  |
|------|-------|--|--|--|--|
| prod | EU-03 |  |  |  |  |
|------|-------|--|--|--|--|

### 3.4 Cost estimation template

| Input                               | Low scenario | Medium scenario | High scenario |
|-------------------------------------|--------------|-----------------|---------------|
| Users                               | 25           | 100             | 500           |
| Interactions per user per day       | 3            | 5               | 10            |
| Share of follow-up interactions     | 30%          | 50%             | 70%           |
| Model calls per initial interaction | 2            | 3               | 5             |
| Model calls per follow-up           | 1            | 2               | 4             |
| Average prompt size                 | 2k tokens    | 6k tokens       | 15k tokens    |
| Average answer size                 | 300 tokens   | 700 tokens      | 1.5k tokens   |
| Query executions per interaction    | 0.8          | 1.2             | 2.0           |
| Average query execution cost        | £0.005       | £0.05           | £0.50         |
| Log retention period                | 30 days      | 90 days         | 180+ days     |
| Evaluation runs per month           | 1            | 4               | 10+           |

Estimated monthly cost should include model calls, metadata retrieval, query execution, validation, logging, trace storage, evaluation, regression testing, monitoring and operational support.

## 4 Evaluation, model-switching and performance tracking

### 4.1 Simple model evaluation approach

Model evaluation should compare models on the same task, with the same inputs, against the same expected outputs. The evaluation should include both successful questions and failure cases.

| Evaluation step           | What to do                                                                                         |
|---------------------------|----------------------------------------------------------------------------------------------------|
| Define task               | Evaluate one task at a time: retrieval, query generation, answer generation or follow-up handling. |
| Build test set            | Include common questions, ambiguous questions, edge cases and out-of-scope questions.              |
| Fix inputs                | Use the same metadata, prompts, examples and user context for each model.                          |
| Run blind comparison      | Compare outputs without favouring a provider or model family.                                      |
| Score outputs             | Use agreed dimensions, not personal preference.                                                    |
| Review failures           | Identify whether failure came from retrieval, model reasoning, validation or answer generation.    |
| Estimate cost and latency | Compare quality against cost and response time.                                                    |
| Select by task            | Choose the simplest model pattern that meets quality, safety and cost thresholds.                  |

### 4.2 Evaluation dimensions

| Dimension            | Question to answer                                                         |
|----------------------|----------------------------------------------------------------------------|
| Retrieval quality    | Did the model retrieve the right metadata, metric, dimension and examples? |
| Query correctness    | Did the query use the right source, metric, filters, joins and grain?      |
| Validation behaviour | Did the system block unsafe, invalid or unauthorised queries?              |
| Answer correctness   | Did the final answer match the trusted result?                             |
| Caveat handling      | Were limitations, assumptions and caveats preserved?                       |
| Safe failure         | Did the model clarify, refuse or escalate when needed?                     |
| Follow-up handling   | Did it inherit the right context and avoid scope drift?                    |
| Consistency          | Did repeated runs produce stable results?                                  |
| Latency              | Was the response fast enough for the user journey?                         |
| Cost                 | Was the model pattern sustainable for expected usage?                      |
| Data exposure        | Was unnecessary data, metadata or query detail avoided?                    |
| Maintainability      | Can prompts, models and evaluation sets be updated safely?                 |

### 4.3 Technology patterns for switching and tracking

Teams should avoid hard-coding model choices directly into the orchestration logic. A simple abstraction layer makes it easier to compare models, switch providers, route tasks and track performance over time.

| Technology / pattern          | Where it helps                                                         | Watch-outs                                            |
|-------------------------------|------------------------------------------------------------------------|-------------------------------------------------------|
| Model gateway                 | Centralises provider access, routing, credentials, limits and fallback | Can become a bottleneck if poorly governed            |
| LLM orchestration framework   | Supports model calls, chains, tools, retrieval and tracing             | Avoid framework lock-in or unnecessary complexity     |
| Prompt / model registry       | Tracks prompt versions, model versions and task ownership              | Must be linked to evaluation and change control       |
| Experiment tracking           | Compares model outputs, latency, cost and quality scores               | Needs consistent datasets and scoring rules           |
| Observability platform        | Tracks traces, errors, costs, latency and user feedback                | Sensitive prompts and results need retention controls |
| Feature flags / routing rules | Allows controlled rollout of model or prompt changes                   | Requires rollback and approval process                |
| Evaluation harness            | Runs repeatable tests across models and prompts                        | Must include edge cases and failure scenarios         |

Example technologies may include LangChain, LlamaIndex, Semantic Kernel, Haystack, DSPy, MLflow, Weights & Biases, LangSmith, Helicone, PromptLayer, OpenTelemetry, provider-native tracing and cloud-native monitoring. The specific choice matters less than the capability: the team needs a controlled way to switch models, compare performance, trace failures and govern changes.

## 5 Activities

### 5.1 Confirm orchestration principles and constraints

For model, cost and evaluation assumptions that cut across activities, see Sections 2 to 4.

#### 5.1.1 Orchestration principles checklist

| Principle                      | Design question                                                            | Evidence to capture                                            |
|--------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------|
| Governed context first         | Does the model use approved metadata before generating queries or answers? | Metadata sources, retrieval rules, grounding approach.         |
| Model judgement is constrained | What must not be decided by the model alone?                               | Rules for metrics, joins, filters, access and caveats.         |
| Tool use is bounded            | Which tools can be called, and under what limits?                          | Tool list, permissions, limits and blocked actions.            |
| Validate before action         | What is checked before query execution or response generation?             | Validation rules, dry-run approach, refusal criteria.          |
| Fail safely                    | When should the system clarify, refuse or escalate?                        | Safe-failure rules and examples.                               |
| Explain the answer             | What must be visible to support trust?                                     | Sources, definitions, caveats, assumptions and query evidence. |
| Log what matters               | Can failures, misuse and quality issues be investigated?                   | Required logs, traces and retention assumptions.               |
| Govern change                  | Who approves changes to models, prompts, tools and validation?             | Change owners, approval route and decision log.                |

#### 5.1.2 Security, exposure, cost and scale assumptions template

| Area                  | Question                                                                                                 |
|-----------------------|----------------------------------------------------------------------------------------------------------|
| User identity         | How is the user identified?                                                                              |
| Access enforcement    | Where are permissions enforced?                                                                          |
| Row-level security    | Are row-level rules inherited or recreated?                                                              |
| Column restrictions   | Which fields must be hidden, masked or excluded?                                                         |
| Sensitive data        | Can personal, financial or regulated data reach the model?                                               |
| Metadata exposure     | Which schema, metric and example metadata can be shared?                                                 |
| Query text            | Can generated SQL or tool calls be sent to the model?                                                    |
| Result data           | Can query results be passed back to the model for explanation?                                           |
| Logs and traces       | What is logged, retained and searchable?                                                                 |
| Audit needs           | What evidence must be available for audit or investigation?                                              |
| Data residency        | Are there location or platform constraints?                                                              |
| Retention             | How long are prompts, responses, logs and feedback retained?                                             |
| Model deployment      | Where can the model run: EU, US, China, private cloud, on-premise or approved SaaS?                      |
| Model type            | Are open, closed, hosted, private or fine-tuned models allowed for this use case?                        |
| Cross-border transfer | Can prompts, metadata, query text, results, logs or feedback leave the required region?                  |
| Cost envelope         | What cost range is acceptable for POC, MVP and pilot?                                                    |
| Cost drivers          | Which costs need active control: model calls, tokens, retrieval, query execution, logging or evaluation? |
|                       |                                                                                                          |

|               |                                                                          |
|---------------|--------------------------------------------------------------------------|
| Usage limits  | Are there limits by user, question, session, query size or time period?  |
| Cost alerts   | What thresholds should trigger review or throttling?                     |
| User scale    | How many users are expected in POC, MVP, pilot and production?           |
| Concurrency   | How many simultaneous sessions or questions should be supported?         |
| Usage pattern | Are questions expected to be ad hoc, daily, peak-period or event-driven? |
| Response time | What latency is acceptable by question type or user group?               |
| Query load    | What query volume can the warehouse, API or semantic layer support?      |
| Throttling    | When should usage be slowed, queued, blocked or escalated?               |

### 5.1.3 Model and tool constraint register

| Constraint area  | Decision to capture                              | Example                                                |
|------------------|--------------------------------------------------|--------------------------------------------------------|
| Model role       | Which tasks can the model perform?               | Intent routing, query drafting, answer explanation.    |
| Model limits     | What must the model not decide?                  | Metric logic, access rights, unsafe joins.             |
| Prompt scope     | What context can be included in prompts?         | Approved metric cards, schema snippets, caveats.       |
| Tool access      | Which tools can the orchestration call?          | Metadata search, query validation, SQL execution.      |
| Tool permissions | What is each tool allowed to do?                 | Read-only query execution with row limits.             |
| Blocked actions  | What is explicitly prohibited?                   | Write-back, unrestricted SQL, cross-domain joins.      |
| Human review     | Which actions require approval or escalation?    | Sensitive data, high-risk answers, failed validation.  |
| Logging          | What must be logged for each model or tool step? | Prompt ID, metadata version, query, validation result. |

### 5.1.4 Change-control decision log

| Change type              | Example                                     | Approval owner                      | Evidence required                      |
|--------------------------|---------------------------------------------|-------------------------------------|----------------------------------------|
| Model change             | Change SQL-generation model.                | AI / solution architect.            | Evaluation comparison and risk review. |
| Prompt change            | Update query-generation prompt.             | AI / solution architect.            | Regression test results.               |
| Tool change              | Add a new execution tool.                   | Security / governance lead.         | Permission review and logging design.  |
| Metadata contract change | Add new fields to metric cards.             | Data / semantic owner.              | Updated register and version history.  |
| Validation change        | Add or relax a validation rule.             | Evaluation owner / governance lead. | Failure analysis and test evidence.    |
| Threshold change         | Change cost, latency or quality thresholds. | Product owner / operating owner.    | Impact assessment.                     |
| Safe-failure change      | Change refusal or escalation rules.         | Product owner / governance lead.    | User and risk review.                  |

### 5.1.5 Minimum evidence before closing Activity 5.1

Before closing this activity, the team should be able to show:

- security, exposure and deployment assumptions;
- cost, latency and scale assumptions;
- model and tool-use constraints;
- logging and audit assumptions;
- change-control expectations;

## 5.2 Question-to-answer flow worksheet

### 5.2.1 End-to-end flow worksheet

The flow is shown sequentially for clarity, but it may loop. Query generation and validation may iterate when the query needs revision, metadata is missing, permissions fail, or the user must clarify the request. The loop should be bounded: after defined attempts or failed checks, the system should refuse, clarify or escalate rather than continue regenerating.

| Step                   | Purpose                                  | Main input                                     | Main output                            | Owner                         | Evidence captured                     |
|------------------------|------------------------------------------|------------------------------------------------|----------------------------------------|-------------------------------|---------------------------------------|
| User question received | Capture the user request and context.    | User question, user identity, session context. | Normalised request.                    | Product / orchestration owner | Question, timestamp, user/session ID. |
| Intent and scope check | Decide whether the question is in scope. | Normalised request, approved scope.            | In-scope, clarify, refuse or escalate. | Orchestration owner           | Intent, scope decision, reason.       |
| Context                | Decide what prior context is relevant.   | Current question,                              | Context to retain or                   | Orchestration                 | Inherited filters, period,            |

|                       |                                                                                              |                                                  |                                               |                               |                                                             |
|-----------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------|-------------------------------|-------------------------------------------------------------|
| handling              |                                                                                              | session history.                                 | discard.                                      | owner                         | metric, entity.                                             |
| Metadata retrieval    | Retrieve governed context.                                                                   | Question, scope, metadata contract.              | Relevant metrics, dimensions, joins, caveats. | Data / semantic owner         | Metadata IDs, versions, retrieval result.                   |
| Query planning        | Decide whether a query is needed.                                                            | Intent, metadata, context.                       | Query plan or no-query response route.        | AI / solution architect       | Query intent, selected sources, assumptions.                |
| Query generation      | Generate a query or tool call using approved metrics, joins, filters and limits              | Query plan, metadata, user context, constraints. | Draft query or tool call.                     | AI / solution architect       | Prompt/version, generated query/tool call.                  |
| Query validation      | Check syntax, scope, permissions, joins, filters, row limits, cost and risk before execution | Draft query, validation rules, permission        | Approve, revise, block, clarify or escalate   | Evaluation / governance owner | Validation checks, failures, revision count, final decision |
| Execution hand-off    | Execute through approved route.                                                              | Validated query or tool call.                    | Query result or execution failure.            | Data / platform owner         | Execution ID, duration, cost, row count.                    |
| Result interpretation | Interpret result against context and caveats.                                                | Query result, metadata, caveats.                 | Interpreted result.                           | Orchestration owner           | Result checks, caveats applied.                             |
| Answer generation     | Produce the user-facing answer.                                                              | Interpreted result, response rules.              | Answer, caveat, refusal or escalation.        | Orchestration owner           | Answer, sources, assumptions, confidence signals.           |
| Feedback and logging  | Capture feedback and trace.                                                                  | Answer, logs, user response.                     | Trace, feedback, improvement signal.          | Operating owner               | Feedback, trace ID, issue category.                         |

### 5.2.2 Flow decision checklist

| Decision area | Question                                                                   | Decision / assumption |
|---------------|----------------------------------------------------------------------------|-----------------------|
| Scope         | What questions are allowed in the first flow?                              |                       |
| Clarification | When should the system ask a clarification question?                       |                       |
| Refusal       | When should the system refuse rather than answer?                          |                       |
| Escalation    | When should a human or analyst review be triggered?                        |                       |
| Context       | Which parts of prior conversation can be reused?                           |                       |
| Metadata      | Which governed artefacts must be retrieved before query generation?        |                       |
| Query         | When is query generation allowed?                                          |                       |
| Validation    | Which checks must pass before execution?                                   |                       |
| Execution     | Which route is used for execution: SQL, API, semantic layer or BI tool?    |                       |
| Answer        | What must be shown with the answer: source, caveat, period, metric or SQL? |                       |
| Logging       | What must be logged at each step?                                          |                       |
| Feedback      | How can users flag wrong, unclear or unhelpful answers?                    |                       |

### 5.2.3 Safe-failure examples

| Situation                           | Preferred behaviour                                      | Evidence to capture                           |
|-------------------------------------|----------------------------------------------------------|-----------------------------------------------|
| User asks for an unsupported metric | Refuse or explain scope; suggest supported alternatives. | Question, unsupported metric, refusal reason. |
| User question is ambiguous          | Ask a clarification question before querying.            | Ambiguity type, clarification asked.          |
| Metadata retrieval fails            | Do not generate query; escalate or ask for support.      | Retrieval failure, missing artefact.          |
| User lacks permission               | Refuse or return permitted aggregate only.               | Permission check, denied scope.               |
| Query validation fails              | Block execution and explain limitation if appropriate.   | Failed rule, query version, owner.            |
| Result conflicts with caveat        | Answer with caveat or escalate for review.               | Caveat applied, result issue.                 |
| Follow-up changes scope silently    | Restate assumptions or ask clarification.                | Prior context, new context, decision.         |

### 5.2.4 Minimum evidence before closing Activity 5.2

Before closing this activity, the team should be able to show:

- an agreed question-to-answer flow;
- main decision points and failure paths;
- clarification, refusal and escalation rules;
- validation and execution sequence;

- logging points across the flow;
- prototype boundaries and simplifications.

## 5.3 Metadata retrieval and grounding worksheet

### 5.3.1 Metadata contract checklist

| Metadata area    | Required fields                                                          | Runtime use                                   |
|------------------|--------------------------------------------------------------------------|-----------------------------------------------|
| Metric           | ID, name, synonyms, definition, formula, owner, status, version, caveats | Select metric, generate query, explain answer |
| Dimension        | ID, name, synonyms, allowed values, hierarchy, owner, status, version    | Filter, group, drill down, clarify            |
| Join rule        | ID, tables, keys, grain, cardinality, allowed use, caveats               | Prevent unsafe joins and duplication          |
| Filter rule      | ID, default filters, exclusions, date logic, mandatory conditions        | Apply standard business logic                 |
| Queryable asset  | ID, table/view/API, owner, status, grain, access rules, freshness        | Select allowed source                         |
| Example question | Question, expected metric, filters, query pattern, answer shape          | Guide query generation and evaluation         |
| Security rule    | Role, row/column restrictions, masking, aggregation limits               | Enforce access and exposure controls          |
| Quality limit    | Freshness, completeness, known gaps, reliability caveats                 | Decide whether to answer, caveat or refuse    |

### 5.3.2 Retrieval design worksheet

Where retrieval or ranking uses a model, the model pattern, cost and evaluation assumptions should align with Sections 2 to 4.

| Design area       | Question                                                     | Decision / assumption |
|-------------------|--------------------------------------------------------------|-----------------------|
| Retrieval source  | Where will metadata be retrieved from?                       |                       |
| Retrieval timing  | Is metadata retrieved at runtime, build time or both?        |                       |
| Search method     | Is retrieval keyword-based, semantic, rules-based or hybrid? |                       |
| Ranking           | How are competing metadata results ranked?                   |                       |
| Versioning        | How is the approved version identified?                      |                       |
| Scope filtering   | How are out-of-scope assets excluded?                        |                       |
| Access filtering  | Is metadata filtered by user permission or role?             |                       |
| Conflict handling | What happens when definitions conflict?                      |                       |
| Missing metadata  | When should the system clarify, refuse or escalate?          |                       |
| Logging           | Which metadata IDs and versions are logged?                  |                       |
| Evaluation        | How is retrieval quality tested?                             |                       |

### 5.3.3 Grounding rules

| Grounding area     | Rule to define                                         | Example decision                                                  |
|--------------------|--------------------------------------------------------|-------------------------------------------------------------------|
| Query generation   | What metadata must be present before query generation? | Metric ID, source asset, grain, join rule and filters required.   |
| Validation         | Which metadata is checked before execution?            | Approved joins, access rules, row limits and mandatory filters.   |
| Answer generation  | Which metadata must appear in the answer or caveat?    | Metric definition, period, caveat and source.                     |
| Follow-up handling | Which metadata can be inherited across turns?          | Metric and period may persist; new dimension requires validation. |
| Refusal            | Which metadata gaps should block an answer?            | No approved metric or unsafe join path.                           |

### 5.3.4 Missing or conflicting metadata handling

| Situation                  | Preferred behaviour                             | Evidence to capture                    |
|----------------------------|-------------------------------------------------|----------------------------------------|
| Metric not found           | Ask clarification or refuse unsupported metric. | User term, retrieval result, decision. |
| Multiple definitions found | Ask clarification or use approved scope rule.   | Candidate IDs, ranking, chosen rule.   |
| Join rule missing          | Block query and route to remediation.           | Tables, missing join, owner.           |
| Caveat missing             | Do not present answer as fully trusted.         | Metric/source, missing caveat.         |
| Metadata stale             | Warn, refuse or escalate depending on risk.     | Version, date, status.                 |
| Access rule unclear        | Block execution until rule is confirmed.        | User role, asset, unresolved rule.     |

### 5.3.5 Minimum evidence before closing Activity 5.3

Before closing this activity, the team should be able to show:

- required metadata fields for orchestration;
- retrieval source and timing;
- grounding rules for query generation, validation and answers;
- versioning and conflict-handling rules;
- missing-metadata behaviours;
- logging requirements for metadata IDs and versions.

## 5.4 Model, prompt and routing strategy worksheet

### 5.4.1 Model/task allocation

| Task                        | Selected model / pattern | Reason | Constraints | Fallback |
|-----------------------------|--------------------------|--------|-------------|----------|
| Intent routing              |                          |        |             |          |
| Metadata ranking            |                          |        |             |          |
| Query generation            |                          |        |             |          |
| Query validation support    |                          |        |             |          |
| Answer generation           |                          |        |             |          |
| Follow-up handling          |                          |        |             |          |
| Safe-failure classification |                          |        |             |          |

### 5.4.2 Prompt control checklist

| Area              | Question                                        | Decision / assumption |
|-------------------|-------------------------------------------------|-----------------------|
| Prompt ownership  | Who owns each prompt?                           |                       |
| Prompt versioning | How are prompt versions tracked?                |                       |
| Prompt inputs     | What data and metadata can enter the prompt?    |                       |
| Prompt exclusions | What must never be included?                    |                       |
| Prompt testing    | Which regression tests must pass after changes? |                       |
| Prompt logging    | Which prompt ID, version and inputs are logged? |                       |
| Prompt approval   | Who approves changes before release?            |                       |

### 5.4.3 Routing and fallback rules

| Situation          | Preferred route         | Fallback / escalation        |
|--------------------|-------------------------|------------------------------|
| Simple intent      | Rules or small model    | Medium model if unclear      |
| Metadata ambiguity | Retrieval plus ranking  | Ask clarification            |
| Complex SQL        | Medium or strong model  | Escalate if validation fails |
| Validation failure | Deterministic block     | Revise once or escalate      |
| High-risk answer   | Stronger review path    | Human review or refusal      |
| Model unavailable  | Approved fallback model | Degraded mode or pause       |

### 5.4.4 Minimum evidence before closing Activity 5.4

Before closing this activity, the team should be able to show:

- model/task allocation;
- prompt ownership and versioning;
- routing and fallback rules;
- model deployment and retention assumptions;
- cost and latency assumptions;
- evaluation approach for shortlisted models.

## 5.5 Tool boundaries and execution hand-off worksheet

### 5.5.1 Tool registry

| Tool                    | Purpose                              | Allowed actions                             | Prohibited actions                             | Identity model                    | Owner                         |
|-------------------------|--------------------------------------|---------------------------------------------|------------------------------------------------|-----------------------------------|-------------------------------|
| Metadata retrieval tool | Retrieve approved metadata.          | Search/read approved artefacts.             | Edit metadata or retrieve out-of-scope assets. | Service or user-context filtered. | Data / semantic owner         |
| Query validation tool   | Check query safety before execution. | Parse, dry-run, check rules and limits.     | Execute unapproved queries.                    | Service identity.                 | Evaluation / governance owner |
| Query execution tool    | Run approved query or API call.      | Read-only execution against approved route. | Write-back, DDL, unrestricted SQL.             | User or delegated identity.       | Data / platform owner         |
| Logging / trace tool    | Capture orchestration evidence.      | Store trace, decisions, IDs and outcomes.   | Store unnecessary sensitive data.              | Service identity.                 | Operating owner               |
| Feedback tool           | Capture user feedback.               | Record rating, issue type and comments.     | Override answer quality status without review. | User identity.                    | Product / operating owner     |

### 5.5.2 Tool permission worksheet

| Decision area      | Question                                                                     | Decision / assumption |
|--------------------|------------------------------------------------------------------------------|-----------------------|
| Identity           | Does the tool run under user identity, service identity or delegated access? |                       |
| Scope              | Which datasets, metadata, APIs or environments can the tool access?          |                       |
| Access inheritance | Does the tool inherit existing row, column and masking controls?             |                       |
| Input limits       | What input size, query shape or prompt content is allowed?                   |                       |
| Output limits      | What row count, fields, result size or detail level can be returned?         |                       |
| Cost limits        | What scan, token, query or API cost thresholds apply?                        |                       |
| Timeout limits     | When should the tool stop, retry, queue or fail?                             |                       |
| Environment        | Can the tool run in dev, test, pilot and production?                         |                       |
| Logging            | What input, output, decision and error details are logged?                   |                       |
| Escalation         | What failures require human review or support?                               |                       |

### 5.5.3 Execution hand-off pattern

| Step                 | Control question                                        | Required evidence                               |
|----------------------|---------------------------------------------------------|-------------------------------------------------|
| Prepare tool request | Is the request scoped, authorised and complete?         | User/session ID, metadata IDs, selected tool.   |
| Validate input       | Is the input safe, allowed and within limits?           | Input checks, blocked fields, size limits.      |
| Execute tool call    | Is the call routed through the approved execution path? | Tool ID, execution route, timestamp.            |
| Validate output      | Is the result safe to return or pass to the model?      | Row count, masked fields, caveats, risk checks. |
| Handle failure       | Should the system retry, clarify, refuse or escalate?   | Error type, retry count, final decision.        |
| Log trace            | Can the action be reconstructed later?                  | Trace ID, inputs, outputs, decisions, owner.    |

### 5.5.4 Tool failure handling

| Failure                          | Preferred behaviour                                         | Evidence to capture                       |
|----------------------------------|-------------------------------------------------------------|-------------------------------------------|
| Metadata tool returns no result  | Ask clarification, refuse or route to metadata remediation. | Search terms, missing artefact, decision. |
| Validation tool fails            | Block execution and route to review if needed.              | Failed rule, query version, owner.        |
| Execution tool times out         | Retry within limits or explain temporary failure.           | Timeout, retry count, duration.           |
| Execution cost exceeds threshold | Block or require approval.                                  | Estimated cost, threshold, decision.      |
| Result exceeds row limit         | Aggregate, restrict, paginate or refuse.                    | Row count, limit, applied action.         |
| Tool returns restricted fields   | Mask, block or escalate.                                    | Field names, policy rule, decision.       |
| Logging tool fails               | Continue only if minimum audit evidence is preserved.       | Missing log, fallback trace, owner.       |

### 5.5.5 Minimum evidence before closing Activity 5.5

Before closing this activity, the team should be able to show:

- approved tool registry;
- identity and permission model;
- read/write and execution limits;

- timeout, row-limit and cost controls;
- tool failure behaviours;
- logging requirements for tool calls;
- unresolved tool risks and owners.

## 5.6 Query generation and validation worksheet

The examples in this annex are illustrative. They should be adapted to the actual use case, data sensitivity, user scope, technology stack, delivery stage and organisational controls. The aim is to provide a practical starting point, not a mandatory query policy.

### 5.6.1 Applying approved metrics, joins, filters and limits

| Design area        | Concrete example                                                                   | Expected control                                                           |
|--------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Approved metric    | User asks for “revenue”; system maps to approved net_revenue metric.               | Use metric ID, approved formula, owner and caveats from metadata contract. |
| Approved dimension | User asks “by region”; system maps to approved customer or sales region dimension. | Use only approved dimensions for the selected metric.                      |
| Approved join      | Revenue by product uses approved order-line to product join.                       | Reject unapproved joins or many-to-many joins without a rule.              |
| Date logic         | “Last month” maps to approved fiscal or calendar period rule.                      | Apply approved date rule and show period used.                             |
| Standard filters   | Revenue excludes cancelled orders and test accounts.                               | Apply mandatory filters automatically.                                     |
| Row limit          | User asks for customer-level breakdown.                                            | Apply row limit, aggregation threshold or refusal rule.                    |
| Cost limit         | Query scans more data than allowed.                                                | Dry-run, block or require approval.                                        |
| Access rule        | User asks for region they cannot access.                                           | Enforce row/column security before execution.                              |

### 5.6.2 Query type policy

| Query type               | Status     | Example                                 | Expected behaviour                                |
|--------------------------|------------|-----------------------------------------|---------------------------------------------------|
| Simple aggregate         | Allowed    | Revenue last month.                     | Generate, validate and execute.                   |
| Grouped aggregate        | Allowed    | Revenue by region.                      | Execute if dimension is approved.                 |
| Trend query              | Allowed    | Monthly revenue for the last 12 months. | Execute with approved date logic.                 |
| Drill-down               | Restricted | Revenue by customer.                    | Allow only if permissions and row limits pass.    |
| Cross-domain join        | Restricted | Revenue by support ticket category.     | Allow only if approved join path exists.          |
| Free-text SQL request    | Restricted | “Run this SQL.”                         | Validate against policy or refuse.                |
| Forecasting / prediction | Restricted | “Forecast next quarter revenue.”        | Route to supported analytical workflow or refuse. |
| Write / update           | Blocked    | Update customer status.                 | Refuse; T2D should be read-only.                  |
| DDL / destructive SQL    | Blocked    | Drop table, create table, alter view.   | Block deterministically.                          |
| Unbounded detail export  | Blocked    | Export all customer transactions.       | Refuse or route to governed export process.       |

### 5.6.3 Validation checks before execution

| Validation check      | Example failure                                           | Expected action                                                |
|-----------------------|-----------------------------------------------------------|----------------------------------------------------------------|
| Syntax check          | Invalid SQL.                                              | Revise within limits or block.                                 |
| Scope check           | Question outside approved use case.                       | Refuse or escalate.                                            |
| Permission check      | User lacks access to region, product or table.            | Block or restrict result.                                      |
| Source check          | Query uses unapproved table.                              | Block.                                                         |
| Metric check          | Query uses unapproved calculation.                        | Revise or block.                                               |
| Join check            | Query uses unsafe many-to-many join.                      | Block or require approved join rule.                           |
| Filter check          | Mandatory exclusions missing.                             | Revise query.                                                  |
| Grain check           | Aggregation mixes incompatible grains.                    | Block or revise.                                               |
| Row-limit check       | Result may return too many rows.                          | Aggregate, limit or block.                                     |
| Cost check            | Query exceeds scan or compute threshold.                  | Dry-run, block or require approval.                            |
| Sensitive-field check | Query selects restricted fields.                          | Mask, remove or block.                                         |
| Result-shape check    | Output is too detailed for answer use.                    | Aggregate, summarise or block.                                 |
| Runtime estimate      | Query is expected to exceed runtime or timeout threshold. | Block, simplify, ask user to narrow scope or require approval. |

#### 5.6.4 Revise, clarify, refuse or escalate rules

| Situation                   | Preferred action                                   | Example response behaviour                                                                 |
|-----------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------|
| Query syntax fails          | Revise once or twice, then block.                  | "I could not generate a valid query for this request."                                     |
| User term is ambiguous      | Clarify.                                           | "Do you mean gross revenue or net revenue?"                                                |
| Metric not approved         | Refuse or suggest supported metric.                | "I cannot answer using an unapproved revenue definition."                                  |
| Join path missing           | Refuse or escalate to data owner.                  | "This breakdown is not currently supported."                                               |
| Permission fails            | Refuse or return authorised aggregate.             | "You do not have access to that level of detail."                                          |
| Query cost too high         | Simplify, ask user to narrow, or require approval. | "Please narrow the time period or grouping."                                               |
| Result too detailed         | Aggregate or refuse.                               | "I can provide a summary, not row-level records."                                          |
| Validation repeatedly fails | Stop loop and escalate.                            | "This question needs review before it can be answered."                                    |
| Runtime estimate too high   | Ask user to narrow scope or reduce detail.         | "This query may take too long. Please reduce the time period, filters or level of detail." |

#### 5.6.5 Validation approach

| Validation layer                     | Best handled by              | Notes                                                |
|--------------------------------------|------------------------------|------------------------------------------------------|
| Block write / delete / drop commands | Deterministic rules          | Should never rely on model judgement.                |
| Restrict tables and columns          | Deterministic rules          | Use allowlists, denylists and permissions.           |
| Enforce row limits and timeouts      | Deterministic rules          | Apply before execution.                              |
| Check approved metrics and joins     | Rules plus metadata checks   | Use Phase 3 metadata contract.                       |
| Detect ambiguous business terms      | Model-assisted               | Should trigger clarification.                        |
| Review complex SQL intent            | Model-assisted plus rules    | Useful for harder queries, but not sufficient alone. |
| Assess answer caveats                | Model-assisted plus metadata | Must preserve approved caveats.                      |

#### 5.6.6 Minimum evidence before closing Activity 5.6

Before closing this activity, the team should be able to show:

- approved query-generation constraints;
- allowed, restricted and blocked query types;
- validation checks before execution;
- revise, clarify, refuse and escalation rules;
- deterministic controls for obvious unsafe actions;
- logging requirements for query versions and validation results;
- unresolved query risks and owners.

### 5.7 Answer generation and safe-failure worksheet

#### 5.7.1 Answer content checklist

| Answer element    | Required / optional     | Example                                                 | Notes                          |
|-------------------|-------------------------|---------------------------------------------------------|--------------------------------|
| Direct answer     | Required                | "Revenue last month was £4.2m."                         | Should match the query result. |
| Metric definition | Required where relevant | "Revenue means net revenue excluding cancelled orders." | Use approved metadata.         |
| Period used       | Required                | "Period: March 2026."                                   | Avoid hidden date assumptions. |
| Filters applied   | Required where relevant | "Region: UK; channel: online."                          | Show material filters.         |
| Source            | Required                | "Source: governed sales mart."                          | Use approved source name.      |
| Caveats           | Required where relevant | "Excludes late adjustments."                            | Preserve approved caveats.     |
| Assumptions       | Required where relevant | "Assumed calendar month, not fiscal period."            | Clarify if high risk.          |

|                     |          |                                              |                                                     |
|---------------------|----------|----------------------------------------------|-----------------------------------------------------|
| SQL / query         | Optional | “Show SQL.”                                  | Useful for technical users; governed access needed. |
| Confidence signal   | Optional | “Based on approved metric and current data.” | Avoid false precision.                              |
| Suggested follow-up | Optional | “Would you like this by region?”             | Should stay within approved scope.                  |

### 5.7.2 Result interpretation checks

| Check                    | Example issue                           | Expected action                                       |
|--------------------------|-----------------------------------------|-------------------------------------------------------|
| Empty result             | No rows returned.                       | Explain no result or ask user to narrow/change scope. |
| Partial result           | Data missing for one region.            | Caveat or escalate.                                   |
| Stale data               | Data last refreshed 10 days ago.        | Show freshness caveat or refuse if too stale.         |
| Unexpected spike         | Revenue increases 300%.                 | Caveat and suggest validation or analyst review.      |
| Suppressed detail        | Row-level result blocked by policy.     | Return permitted aggregate or refuse.                 |
| Conflicting caveat       | Result conflicts with known limitation. | Apply caveat or escalate.                             |
| Low retrieval confidence | Metric or source match uncertain.       | Clarify before answering.                             |

### 5.7.3 Refusal, clarification and escalation rules

| Situation                 | Preferred behaviour                 | Example response pattern                                  |
|---------------------------|-------------------------------------|-----------------------------------------------------------|
| Ambiguous metric          | Clarify.                            | “Do you mean gross revenue or net revenue?”               |
| Unsupported question      | Refuse and suggest supported scope. | “This question is outside the approved T2D scope.”        |
| Missing approved metric   | Refuse or escalate.                 | “I cannot answer without an approved metric definition.”  |
| Unsafe join or query path | Refuse or route to remediation.     | “This breakdown is not currently supported.”              |
| Permission failure        | Refuse or restrict.                 | “You do not have access to that level of detail.”         |
| High-risk result          | Escalate or caveat.                 | “This result needs review before use in decision-making.” |
| Follow-up changes scope   | Clarify or restate assumptions.     | “Do you want to keep the same region and period?”         |

### 5.7.4 Evidence visibility options

| User type                     | Recommended evidence                                    | Watch-outs                               |
|-------------------------------|---------------------------------------------------------|------------------------------------------|
| Executive user                | Definition, period, source, caveats.                    | Avoid overwhelming detail.               |
| Business manager              | Definition, filters, source, caveats, freshness.        | Keep explanation readable.               |
| Analyst                       | Definition, filters, source, caveats, SQL/query.        | SQL visibility must respect permissions. |
| Auditor / governance reviewer | Metadata IDs, query, validation result, logs, versions. | Usually not shown in normal user answer. |

### 5.7.5 Answer safety checks before response

| Safety check     | Question                                       | Action if failed                |
|------------------|------------------------------------------------|---------------------------------|
| Metric check     | Is the approved metric definition used?        | Block or revise.                |
| Source check     | Is the approved source shown?                  | Add source or block.            |
| Period check     | Is the time period explicit?                   | Add period or clarify.          |
| Caveat check     | Are required caveats preserved?                | Add caveat or block.            |
| Permission check | Does the answer expose restricted information? | Mask, aggregate or refuse.      |
| Certainty check  | Is the wording stronger than the evidence?     | Soften wording or caveat.       |
| Follow-up check  | Are inherited assumptions visible?             | Restate assumptions or clarify. |

### 5.7.6 Minimum evidence before closing Activity 5.7

Before closing this activity, the team should be able to show:

- answer content rules;
- metadata and evidence visibility rules;
- result interpretation checks;
- refusal, clarification and escalation rules;
- answer safety checks;

- SQL/query visibility position;
- logging requirements for answers and refusal reasons.

## 5.8 Multi-turn conversation handling worksheet

### 5.8.1 Context inheritance rules

Context inheritance should not mean passing the whole conversation back to the model. A safer pattern is to maintain a small, structured session state that records only the context needed for follow-up questions. Each new turn should update, reuse, reset or reject that state.

| Context element  | How to implement                                            | Example                                  | Required control                    |
|------------------|-------------------------------------------------------------|------------------------------------------|-------------------------------------|
| Metric           | Store selected metric ID and label in session state.        | metric_id = net_revenue                  | Revalidate if user changes metric.  |
| Time period      | Store explicit period and period type.                      | period = Mar 2026, type = calendar month | Restate period in answer.           |
| Filters          | Store filter fields and values.                             | region = UK, channel = online            | Show material filters.              |
| Dimension        | Store current grouping only if approved.                    | group_by = region                        | Revalidate when dimension changes.  |
| Entity           | Store entity ID or label where needed.                      | customer = ABC Ltd                       | Recheck access before reuse.        |
| Prior result     | Store result reference, not full raw output where possible. | result_id = R12345                       | Check freshness and scope.          |
| Caveats          | Store caveat IDs linked to metadata.                        | caveat_id = late_adjustments_excluded    | Preserve in follow-up answer.       |
| Access context   | Store role/access decision reference.                       | access_profile = sales_uk_manager        | Recheck for new data/detail.        |
| User preference  | Store only low-risk preferences.                            | format = chart                           | Do not store sensitive preferences. |
| Sensitive detail | Avoid storing unless essential.                             | customer-level rows                      | Mask, aggregate or discard.         |

The system should treat context as structured state, not memory. Each follow-up should classify whether it reuses existing context, modifies it, requires new metadata, requires a new query, or must be clarified/refused. Material inherited assumptions should be shown back to the user.

#### 5.8.1.1 Example

| User turn                      | State change                                     | System behaviour                                                   |
|--------------------------------|--------------------------------------------------|--------------------------------------------------------------------|
| "What was revenue last month?" | Set metric = revenue; period = last month.       | Retrieve metadata, query, answer with period and definition.       |
| "Break it down by region."     | Reuse metric and period; add dimension = region. | Validate region dimension and re-query.                            |
| "What about margin?"           | Metric change detected.                          | Retrieve margin metadata or clarify if multiple definitions exist. |
| "Show customer-level detail."  | Detail level change detected.                    | Recheck permissions and row/detail limits.                         |

### 5.8.2 Follow-up handling patterns

| Follow-up type         | Example                              | Preferred behaviour                                                     |
|------------------------|--------------------------------------|-------------------------------------------------------------------------|
| Drill-down             | "Break that down by region."         | Reuse metric and period, validate dimension, re-query.                  |
| Filter change          | "Only for the UK."                   | Apply new filter, restate assumptions, re-query.                        |
| Period change          | "What about last quarter?"           | Replace period, re-query.                                               |
| Comparison             | "Compare that to last year."         | Keep metric, add comparison period, re-query.                           |
| Explanation            | "Why did it change?"                 | Use supported drivers only; clarify if root-cause data is not approved. |
| Format change          | "Show as a chart."                   | Reuse result if still valid.                                            |
| Scope expansion        | "Now include customer-level detail." | Check permission and row/detail limits.                                 |
| Ambiguous follow-up    | "What about margin?"                 | Clarify metric or scope before answering.                               |
| Out-of-scope follow-up | "Forecast next quarter."             | Refuse or route to supported workflow.                                  |

### 5.8.3 Clarify, re-query, reuse or refuse rules

| Situation                              | Preferred action                        | Reason                                         |
|----------------------------------------|-----------------------------------------|------------------------------------------------|
| Follow-up keeps same metric and period | Reuse or re-query depending on request. | Avoid unnecessary cost but keep accuracy.      |
| Follow-up changes metric               | Clarify or retrieve new metadata.       | Metric logic may differ.                       |
| Follow-up changes dimension            | Revalidate and usually re-query.        | Dimensions may have different joins or limits. |

|                                        |                                     |                                        |
|----------------------------------------|-------------------------------------|----------------------------------------|
| Follow-up adds restricted detail       | Check permissions before answering. | Prevent exposure through conversation. |
| Prior result is stale or partial       | Re-query or caveat.                 | Avoid answering from weak evidence.    |
| Context is ambiguous                   | Clarify.                            | Prevent hidden assumptions.            |
| Follow-up moves outside approved scope | Refuse or escalate.                 | Keep system within governed use case.  |
| Multiple failed turns occur            | Escalate or end safely.             | Avoid repeated guessing.               |

#### 5.8.4 Session retention and logging

| Area               | Design question                              | Decision / assumption |
|--------------------|----------------------------------------------|-----------------------|
| Session duration   | How long is conversation context retained?   |                       |
| Context summary    | Is full history retained or summarised?      |                       |
| Sensitive context  | What must not be retained across turns?      |                       |
| User identity      | How is identity linked to the session?       |                       |
| Permission recheck | When are permissions revalidated?            |                       |
| Context log        | Which inherited assumptions are logged?      |                       |
| Refusal log        | Are refusals and reasons captured?           |                       |
| Feedback link      | Can feedback be linked to the relevant turn? |                       |
| Retention policy   | How long are conversation traces retained?   |                       |

#### 5.8.5 Conversation safety checks

| Safety check     | Question                                      | Action if failed           |
|------------------|-----------------------------------------------|----------------------------|
| Scope check      | Is the follow-up still in approved scope?     | Refuse or escalate.        |
| Context check    | Are inherited assumptions still valid?        | Restate, clarify or reset. |
| Permission check | Does the user still have access?              | Block, mask or aggregate.  |
| Freshness check  | Is the prior result still usable?             | Re-query or caveat.        |
| Detail check     | Does the follow-up request restricted detail? | Refuse or restrict.        |
| Caveat check     | Are prior caveats still preserved?            | Restate caveats.           |
| Ambiguity check  | Could the follow-up mean several things?      | Clarify.                   |

#### 5.8.6 Minimum evidence before closing Activity 5.8

Before closing this activity, the team should be able to show:

- context inheritance rules;
- follow-up handling patterns;
- clarify, re-query, reuse and refusal rules;
- session retention assumptions;
- conversation safety checks;
- logging requirements for context and turn-level decisions.

### 5.9 Model and orchestration evaluation worksheet

#### 5.9.1 Evaluation tool options

| Tool / pattern        | Where it helps                                                               | Pros                                               | Cons                                                 |
|-----------------------|------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------|
| Spreadsheet scorecard | Early manual evaluation of model outputs, answers and refusals.              | Simple, fast, accessible to business reviewers.    | Hard to scale, weak traceability, manual effort.     |
| SQL / BI comparison   | Compare generated results against trusted reports or benchmark queries.      | Strong for numeric correctness and reconciliation. | Does not assess explanation quality or safe failure. |
| Golden test set       | Repeatable set of approved questions, expected queries and expected answers. | Good baseline for regression and model comparison. | Needs maintenance as metrics, data and scope change. |
| Human review panel    | SMEs review answer usefulness, caveats and business interpretation.          | Captures judgement and trust issues.               | Slow, subjective, needs clear scoring rubric.        |
| Automated unit tests  | Test deterministic rules, blocked SQL, permissions and validation gates.     | Fast, repeatable, CI/CD friendly.                  | Does not cover nuanced language or judgement.        |
| LLM-as-judge          | Model reviews outputs against rubric or reference.                           | Useful for scale and qualitative.                  | Must be validated; can introduce bias.               |

|                                  |                                                                |                                            |                                                            |
|----------------------------------|----------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------|
| LLM-as-judge                     | Model reviews outputs against rubric or reference answer.      | Useful for scale and qualitative checks.   | Must be validated, can introduce bias or false confidence. |
| Prompt / model registry          | Tracks prompt versions, model versions and evaluation results. | Supports governance and rollback.          | Needs discipline and ownership.                            |
| Experiment tracking              | Compares models, prompts, cost, latency and scores.            | Good for structured model selection.       | Requires consistent datasets and metrics.                  |
| Observability / tracing platform | Tracks traces, failures, latency, cost and feedback.           | Essential for debugging and monitoring.    | Sensitive prompts/results need retention controls.         |
| Evaluation framework             | Runs repeatable evals across models, prompts and datasets.     | Useful for regression and model switching. | Can be over-engineered for early POCs.                     |

### 5.9.2 Example technologies

| Technology            | Likely use                                                               | Pros                                                    | Cons                                                    |
|-----------------------|--------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| LangSmith             | Tracing, evaluation, dataset management for LangChain-based flows.       | Strong tracing and prompt/model comparison.             | Most useful if using LangChain ecosystem.               |
| LangChain             | Orchestration, tool calling, retrieval and evaluation hooks.             | Broad ecosystem and quick prototyping.                  | Can add complexity and framework lock-in.               |
| LlamaIndex            | Retrieval, metadata grounding and evaluation around data connectors.     | Strong for retrieval-heavy workflows.                   | Less suited if orchestration is not retrieval-centred.  |
| Semantic Kernel       | Enterprise-oriented orchestration, especially in Microsoft environments. | Fits Azure / Microsoft stack well.                      | Less neutral if not using Microsoft ecosystem.          |
| MLflow                | Experiment tracking, model registry and evaluation logging.              | Familiar MLOps pattern, strong governance fit.          | Less specialised for LLM traces and conversations.      |
| Weights & Biases      | Experiment tracking, evaluation and model comparison.                    | Strong experiment management.                           | May be heavier than needed for business POCs.           |
| Ragas                 | Retrieval and RAG evaluation.                                            | Useful for retrieval quality and grounding.             | Needs careful adaptation for T2D, not generic RAG only. |
| DeepEval              | LLM evaluation tests and metrics.                                        | Good for automated test suites.                         | Metrics must be customised to T2D risks.                |
| Promptfoo             | Prompt and model regression testing.                                     | Lightweight and practical for comparing prompts/models. | Less complete for full production observability.        |
| OpenTelemetry         | Standardised tracing across systems.                                     | Vendor-neutral and production-friendly.                 | Requires engineering setup.                             |
| Helicone              | LLM observability, cost and latency tracking.                            | Quick visibility into model calls.                      | Depends on deployment and routing pattern.              |
| Provider-native tools | OpenAI, Anthropic, Azure, Google or AWS tracing/evaluation tools.        | Integrated with provider stack.                         | May increase provider lock-in.                          |

### 5.9.3 Evaluation design template

| Area             | Decision / question                | Example                                                          |
|------------------|------------------------------------|------------------------------------------------------------------|
| Evaluation scope | Which tasks are evaluated?         | Retrieval, SQL generation, answer generation, follow-ups.        |
| Test set         | Which questions are included?      | Common, ambiguous, edge-case and out-of-scope questions.         |
| Reference answer | What is the expected answer?       | Trusted SQL result, approved caveats, expected refusal.          |
| Scoring method   | How is quality scored?             | 1-5 scale with pass/fail blockers.                               |
| Human review     | Who reviews business quality?      | SME, data owner, product owner.                                  |
| Automated checks | Which checks are automated?        | SQL validity, blocked commands, row limits, source use.          |
| LLM judge use    | Where is model review acceptable?  | Caveat preservation, answer clarity, refusal quality.            |
| Thresholds       | What score is required to proceed? | No critical failure; minimum score by task.                      |
| Regression       | When are tests rerun?              | Prompt, model, metadata or validation changes.                   |
| Evidence         | What is retained?                  | Test ID, prompt version, model version, output, score, reviewer. |

### 5.9.4 Suggested scoring dimensions

| Dimension            | What to score                                     | Blocking failure example                |
|----------------------|---------------------------------------------------|-----------------------------------------|
| Retrieval quality    | Correct metadata, metric, source, examples.       | Wrong metric retrieved.                 |
| Query correctness    | Correct source, joins, filters, grain and period. | Unsafe join reaches execution.          |
| Validation behaviour | Unsafe or invalid actions are blocked.            | Write/delete/drop query is allowed.     |
| Answer correctness   | Final answer matches trusted result.              | Wrong number or unsupported conclusion. |
| Caveat handling      | Caveats and assumptions are preserved.            | Required caveat omitted.                |
| Safe failure         | Clarifies, refuses or escalates when needed.      | Assumes unsupported question.           |

|                     |                                               |                                              |
|---------------------|-----------------------------------------------|----------------------------------------------|
| Safe failure        | Clarifies, refuses or escalates when needed.  | Answers unsupported question.                |
| Follow-up handling  | Correct context inheritance and revalidation. | User moves outside scope silently.           |
| Security / exposure | Avoids unnecessary data exposure.             | Sensitive data included in prompt or answer. |
| Latency             | Response time fits user journey.              | Answer too slow for intended use.            |
| Cost                | Cost fits usage assumptions.                  | Uses strongest model for every step.         |
| Usability           | Answer is understandable and useful.          | Technically correct but unclear to user.     |

### 5.9.5 Tool selection guidance

| Situation       | Preferred approach                                                                                     |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Early POC       | Spreadsheet scorecard, golden test set, manual SME review, provider logs.                              |
| MVP             | Golden test set, automated validation tests, prompt/model registry, tracing platform.                  |
| Pilot           | Evaluation framework, observability platform, regression tests, feedback-linked monitoring.            |
| Production path | CI/CD evaluation gates, model registry, OpenTelemetry or equivalent tracing, formal approval workflow. |

### 5.9.6 Minimum evidence before closing Activity 5.9

Before closing this activity, the team should be able to show:

- evaluation scope and test set;
- scoring dimensions and thresholds;
- model and prompt comparison approach;
- tooling selected for evaluation and tracing;
- human and automated review responsibilities;
- regression trigger rules;
- retained evaluation evidence and owners.

## 5.10 Quality monitoring and improvement loop worksheet

### 5.10.1 Quality signal catalogue

| Signal             | What it shows                                      | Example metric                                      |
|--------------------|----------------------------------------------------|-----------------------------------------------------|
| User feedback      | Whether users trust or find the answer useful.     | Thumbs down rate by question type.                  |
| Answer correction  | Where answers were wrong or incomplete.            | Number of corrected answers per week.               |
| Failed validation  | Where generated queries or tool calls are blocked. | Validation failure rate by rule.                    |
| Refusals           | Where the system cannot or should not answer.      | Refusal rate by reason.                             |
| Clarifications     | Where user questions are ambiguous.                | Clarification rate by metric or domain.             |
| Escalations        | Where human review is needed.                      | Escalations by risk category.                       |
| Retrieval failures | Where metadata grounding is weak.                  | Missing or conflicting metadata rate.               |
| Query failures     | Where execution fails or times out.                | Query failure and timeout rate.                     |
| Latency            | Whether response time fits the user journey.       | Median and 95th percentile response time.           |
| Cost               | Whether usage is financially sustainable.          | Cost per interaction and cost per active user.      |
| Drift              | Whether quality changes over time.                 | Regression pass rate after model or prompt changes. |
| Usage              | Whether users are adopting the capability.         | Active users, sessions and repeat usage.            |

### 5.10.2 Feedback capture design

| Feedback type      | Capture method                                                     | Use                                         |
|--------------------|--------------------------------------------------------------------|---------------------------------------------|
| Rating             | Thumbs up / down or simple score.                                  | Identify weak answers and trends.           |
| Reason category    | Wrong number, unclear answer, missing caveat, too slow, no answer. | Route issues to the right owner.            |
| Free-text comment  | Optional user explanation.                                         | Understand edge cases and user frustration. |
| Correction         | User or SME provides corrected answer.                             | Add to evaluation set or backlog.           |
| Escalation request | User asks for analyst or owner review.                             | Trigger support or governance route.        |
| Follow-up pattern  | Repeated rephrasing or clarification.                              | Identify unclear answers or poor grounding. |

### 5.10.3 Monitoring dashboard outline

| Area                  | Example measures                                                       | Owner                         |
|-----------------------|------------------------------------------------------------------------|-------------------------------|
| Adoption              | Active users, sessions, interactions per user, repeat usage.           | Product owner                 |
| Answer quality        | Feedback score, correction rate, failed evaluation cases.              | Evaluation owner              |
| Retrieval quality     | Retrieval failures, conflicting metadata, missing artefacts.           | Data / semantic owner         |
| Query safety          | Validation failures, blocked queries, unsafe query attempts.           | Evaluation / governance owner |
| Security and exposure | Permission failures, blocked sensitive fields, audit exceptions.       | Security / governance lead    |
| Performance           | Latency, timeout rate, query duration, model-call duration.            | AI / platform owner           |
| Cost                  | Cost per interaction, model spend, query execution cost, logging cost. | Product / operating owner     |
| Operations            | Incidents, escalations, unresolved issues, SLA breaches.               | Operating owner               |

### 5.10.4 Issue triage and routing

| Issue type              | Likely owner                  | Typical action                                         |
|-------------------------|-------------------------------|--------------------------------------------------------|
| Wrong metric selected   | Data / semantic owner         | Fix metadata, synonyms or retrieval rules.             |
| Wrong SQL / query       | AI / solution architect       | Improve query generation or validation.                |
| Unsafe query blocked    | Evaluation / governance owner | Review rule and confirm expected behaviour.            |
| Missing caveat          | Data / semantic owner         | Update metadata and answer rules.                      |
| Permission issue        | Security / governance lead    | Fix access rule, identity mapping or masking.          |
| Slow answer             | AI / platform owner           | Optimise model, retrieval, query or caching.           |
| High cost               | Product / operating owner     | Adjust routing, limits, caching or usage controls.     |
| Poor user understanding | Product owner / business SME  | Improve answer format or guidance.                     |
| Repeated refusal        | Product owner / data owner    | Decide whether scope should expand or stay restricted. |

### 5.10.5 Improvement triggers

| Trigger                  | Example threshold                                    | Response                                   |
|--------------------------|------------------------------------------------------|--------------------------------------------|
| Low feedback score       | Feedback below agreed threshold for a question type. | Review answers and update backlog.         |
| High correction rate     | Repeated SME corrections for same metric or domain.  | Add regression tests and fix root cause.   |
| Retrieval failure spike  | Missing metadata increases after source change.      | Review metadata contract and indexes.      |
| Validation failure spike | Many queries blocked for same rule.                  | Check prompt, metadata or rule design.     |
| Refusal spike            | Sudden increase in unsupported questions.            | Review scope, user guidance or backlog.    |
| Latency breach           | 95th percentile above target.                        | Optimise routing, query design or caching. |
| Cost breach              | Cost per interaction above threshold.                | Adjust model routing or query limits.      |
| Security event           | Restricted data appears in prompt, log or answer.    | Stop affected route and escalate.          |
| Regression failure       | Golden test set falls below threshold.               | Block release or roll back change.         |

### 5.10.6 Improvement backlog

| Backlog item | Source signal                                      | Root cause | Owner | Priority | Target action |
|--------------|----------------------------------------------------|------------|-------|----------|---------------|
|              | User feedback / validation / regression / incident |            |       |          |               |
|              | User feedback / validation / regression / incident |            |       |          |               |
|              | User feedback / validation / regression / incident |            |       |          |               |

### 5.10.7 Minimum evidence before closing Activity 5.10

- Before closing this activity, the team should be able to show:
- agreed quality signals;
- feedback capture design;
- monitoring dashboard outline;
- issue triage and ownership model;
- improvement triggers and thresholds;

- backlog and remediation process;
- governance link for model, prompt, metadata and validation changes.

## 5.11 Orchestration governance and change-control worksheet

### 5.11.1 5.11.1. Change-control scope

| Change area         | Example change                             | Approval owner                | Evidence required                                     |
|---------------------|--------------------------------------------|-------------------------------|-------------------------------------------------------|
| Model               | Change SQL-generation model.               | AI / solution architect       | Evaluation results, latency/cost impact, risk review. |
| Prompt              | Update answer-generation prompt.           | AI / solution architect       | Regression results, prompt version, approval note.    |
| Tool                | Add a new execution tool.                  | Security / governance lead    | Permission review, logging design, failure behaviour. |
| Metadata contract   | Add new fields to metric cards.            | Data / semantic owner         | Updated contract, version history, downstream impact. |
| Validation rule     | Relax row-limit or join rule.              | Evaluation / governance owner | Failure analysis, risk review, test evidence.         |
| Threshold           | Change cost, latency or quality threshold. | Product / operating owner     | Impact assessment and monitoring plan.                |
| Safe-failure rule   | Change refusal or escalation rule.         | Product / governance owner    | User impact review and risk assessment.               |
| Logging / retention | Change trace or prompt retention.          | Security / governance lead    | Privacy, audit and retention review.                  |

### 5.11.2 5.11.2. Change classification

| Change type                 | Risk level    | Example                                         | Required route                                 |
|-----------------------------|---------------|-------------------------------------------------|------------------------------------------------|
| Minor configuration         | Low           | Update non-material wording in answer template. | Record change and test affected examples.      |
| Prompt or routing change    | Medium        | Change model used for metadata ranking.         | Regression test and owner approval.            |
| Validation change           | Medium / high | Change join, row-limit or cost rule.            | Risk review, test evidence and approval.       |
| Tool permission change      | High          | Allow a tool to access new data source.         | Security review and formal approval.           |
| Security / retention change | High          | Store prompts for longer period.                | Privacy, security and governance review.       |
| Scope expansion             | High          | Add new domain, metric group or user group.     | Product, data, security and evaluation review. |

### 5.11.3 5.11.3. Release evidence checklist

| Evidence item                 | Required for                                          | Owner                      |
|-------------------------------|-------------------------------------------------------|----------------------------|
| Change description            | All changes                                           | Change owner               |
| Reason for change             | All changes                                           | Change owner               |
| Impacted flow step            | All changes                                           | AI / solution architect    |
| Prompt / model / tool version | Relevant technical changes                            | AI / solution architect    |
| Regression test result        | Prompt, model, tool, validation and metadata changes  | Evaluation owner           |
| Security review               | Tool, access, exposure, retention and logging changes | Security / governance lead |
| Cost and latency impact       | Model, routing, query and tool changes                | Product / operating owner  |
| Rollback plan                 | Medium and high-risk changes                          | AI / solution architect    |
| Approval record               | Medium and high-risk changes                          | Governance owner           |

### 5.11.4 5.11.4. Regression and rollback triggers

| Trigger                  | Example                                           | Required response                              |
|--------------------------|---------------------------------------------------|------------------------------------------------|
| Answer quality drop      | Golden test set score falls below threshold.      | Block release or roll back.                    |
| Safe-failure failure     | Unsupported question is answered.                 | Stop affected route and fix rule/prompt.       |
| Security issue           | Restricted data appears in prompt, log or answer. | Disable route and escalate.                    |
| Cost spike               | Cost per interaction exceeds threshold.           | Review routing, model choice and query limits. |
| Latency breach           | Response time exceeds user-journey target.        | Optimise, restrict or roll back.               |
| Validation failure spike | More queries blocked or failing after change.     | Review prompt, metadata or validation rule.    |
| User feedback issue      | Increase in wrong or unclear answer reports.      | Triage and add regression cases.               |

### 5.11.5 5.11.5. Governance cadence

| Cadence | Purpose | Participants |
|---------|---------|--------------|
|         |         |              |

|                          |                                                                    |                                                      |
|--------------------------|--------------------------------------------------------------------|------------------------------------------------------|
| Weekly during prototype  | Review issues, failed tests, design changes and risks.             | Product, AI architect, data owner, evaluation owner. |
| Fortnightly during pilot | Review quality, usage, cost, feedback and change backlog.          | Product, operating owner, governance, business SME.  |
| Monthly after launch     | Review performance, incidents, changes and improvement priorities. | Product, data, AI, security and operating owners.    |
| Ad hoc                   | Review high-risk changes, incidents or security events.            | Relevant decision owners and approvers.              |

5.11.6 5.11.6. Minimum evidence before closing Activity 5.11

Before closing this activity, the team should be able to show:

- defined change-control scope;
- approval owners by change type;
- release evidence requirements;
- regression and rollback triggers;
- governance cadence;
- decision and approval logging route;
- unresolved governance risks and owners.